SALES DATA

SQL Analysis Project

Global Sales Dataset | 2010–2024 | African Market Focus by KN Kgetjepe

14 Years	4 X-Models	2 Analyses	1 Market
Of Data	Analyzed	Core Studies	Africa

01 | About This Project

This project documents a professional SQL data analysis performed on BMW's global sales data set spanning from 2010 to 2024. The analysis focuses on two critical business questions every automotive company needs to answer — how much revenue are we generating, and how sensitive are our customers to price changes?

Using SQL techniques including window functions, stored procedures, and calculated fields, we extract actionable insights that directly inform BMW's pricing strategy and market investment decisions — with a specific focus on the African market.

Business Problem

BMW's African regional sales division reported a sustained decline in X-class vehicle sales between 2010 and 2024. Regional executives required a data-driven SQL investigation to determine whether declining sales were caused by pricing decisions, market conditions, or model-specific under-performance and to recommend a corrective data strategy.

Key Questions This Analysis Answers

1. Which BMW model is generating the most revenue in Africa?
2. Have price increases contributed to the sales volume decline?
3. Are African buyers price sensitive or driven by brand prestige?
4. Which model should BMW prioritize to reverse the sales decline?

Project Scope

Dataset	`bmw sales data (2010-2024)`
Region Filter	Africa
Models in Scope	M(3/5) & I(3/8) & X(1/3/5/6) & (3,5,7)SERIES
Time Period	2010 – 2024 (14 years)
Key Columns	model, year, region, price_usd, sales_volume, sales_classification, transmission, fuel_type
Tool	MySQL Workbench

SQL Techniques Used

SELECT & Filtering	Precise data extraction with WHERE clauses and column aliases
Window Functions	LAG() with PARTITION BY for year-over-year comparisons
Stored Procedures	Reusable query logic with DROP, CREATE, BEGIN/END blocks
Calculated Fields	Derived metrics: total_revenue, price_change_pct, volume_change_pct
Aggregation	SUM(), GROUP BY for executive summary reporting
Data Formatting	ROUND() for clean, presentation-ready numeric output

02 | Query #001 — BMW Vehicle Model Stored Procedure

A stored procedure that encapsulates multiple SELECT queries for BMW's Car mode lineup . Instead of running separate repetitive queries, analysts can easily execute a single CALL statement that returns any required table result by model instantly.

```sql
-- GROUPED DIFFERANT TABLES BY FILTERING THE CAR CLASS MODEL AND RANGE FOR BMW 2010-2024

-- BMW SERIES CLASS TABLES
drop procedure if exists bmw_series_class;
delimiter //
create procedure bmw_series_class ()
begin
select* from bmw.`bmw sales data (2010-2024)` where model = '3 Series';
select* from bmw.`bmw sales data (2010-2024)` where model = '5 Series';
select* from bmw.`bmw sales data (2010-2024)` where model = '7 Series';
end //
delimiter ;
call bmw_series_class();
```

```sql

-- BMW M TABLE
drop procedure if exists bmw_M_class;
delimiter //
create procedure bmw_M_class ()
begin
select* from bmw.`bmw sales data (2010-2024)` where model = 'M5';
select* from bmw.`bmw sales data (2010-2024)` where model = 'M3';
end //
delimiter ;
call bmw_M_class ()
```

Set limit for number of rows returned by queries.
Workbench will automatically add the LIMIT clause with the configured number of r

```sql

-- BMW I CLASS TABLES
drop procedure if exists bmw_I_class;
delimiter //
create procedure bmw_I_class ()
begin
select* from bmw.`bmw sales data (2010-2024)` where model = 'I3';
select* from bmw.`bmw sales data (2010-2024)` where model = 'I8';
end //
delimiter ;
call bmw_I_class ();
```

- Helps analyze a tables set by different products which makes it easier for analyst to select any table sets by vehicle models.
- Time is saved by using call to work with data instead of using select to retrieve data.
- Easier to update data or make the changes on the right data which is also most suitable for consistency.
- It is secured to use because use can be granted without giving full access which helps protect raw data.

```sql
-- BMW X CLASS TABLES
drop procedure if exists bmw_x_class ;
delimiter //
create procedure bmw_x_class()
begin
select* from bmw.`bmw sales data (2010-2024)` where model = 'X1';
select* from bmw.`bmw sales data (2010-2024)` where model = 'X3';
select* from bmw.`bmw sales data (2010-2024)` where model = 'X5';
select* from bmw.`bmw sales data (2010-2024)` where model = 'X6';
end //
delimiter ;
call bmw_x_class();
```

03 | Query #002 — Africa Revenue Trend Analysis

This query pulls all BMW Africa sales data and adds a derived total_revenue column calculated by multiplying price by sales volume. Results are sorted chronologically by year, then by sales classification and volume — giving executives a ranked performance report for the African market.

```sql
-- BMW Sales Revenue for each region 2010-2024
select *,(price_usd*sales_volume) as total_revenue
from bmw.`bmw sales data (2010-2024)`
where region = 'Africa'
order by year,total_revenue
;
-- Revenue Table on Vehicles with Manual transmitions in Africa
select *,(price_usd*sales_volume) as total_revenue
from bmw.`bmw sales data (2010-2024)`
where region = 'Africa'
having transmission = 'manual'
order by year,total_revenue;
```

```sql
-- Revenue Table on Vehicles with Manual transmitions and Hybrid fuel type in Africa
select *,(price_usd*sales_volume) as total_revenue
from bmw.`bmw sales data (2010-2024)`
where region = 'Africa'
having transmission = 'manual' and fuel_type = 'hybrid'
order by year,total_revenue;
```

- Helps analyze which model / year generated the most revenue which can also highlight business demand.

- Helps analyze how revenue has grown and declined over the year for Africa.
- Helps identify which model to focus on for marketing strategies to drive sales on each region.

04 | Query #003 — Price Elasticity of Demand

The most advanced query in this project. Using the LAG() window function partitioned by model, we compare each year's price and sales volume to the previous year — calculating percentage changes that reveal how price-sensitive BMW Africa buyers are across each vehicle model.

```sql

-- Price Elasticity of Demand for BMW X-Class Models (2010-2024)
SELECT
    model,
    year,
    price_usd,
    sales_volume,
    lag(price_usd) over (partition by model order by year) AS prev_price,
    lag(sales_volume) over (partition by model order by year) as prev_sales_volume,
    ROUND(((price_usd - lag(price_usd) OVER (partition by model order by year))
        / lag(price_usd) over (partition by model order by year)) * 100, 2) as price_change_percentage,
    ROUND(((sales_volume - lag(sales_volume) over (partition by model order by year))
        / lag(sales_volume) over (partition by model order by year)) * 100, 2) as volume_change_percentage
from bmw.`bmw sales data (2010-2024)`
where Region = 'Africa'
order by model, year ASC;

```

- Helps identify price sensitivity effect between the price change and volume change for each Region
- Helps know which models are price sensitive vs prestige driven where buyers are less discouraged by high prices
- Guiding future pricing strategy — whether to discount, hold, or premium price each model

05 | How to Read the Results

When the Price Elasticity query runs, results fall into four scenarios. Here's how to interpret each one and what action BMW should take:

#	Price Change	Volume Change	Meaning	BMW Action
1	Price UP	Volume DOWN	Customers are price sensitive	Stop raising prices in Africa

2	Price UP	Volume UP	Prestige effect — luxury signal	Maintain premium pricing strategy
3	Price DOWN	Volume UP	Discounting is driving demand	Run more promotions to boost volume
4	Price DOWN	Volume DOWN	External factors suppressing sales	Investigate competition & economy

07 | Business Recommendations

Freeze Price Increases

African buyers show strong price sensitivity — especially post-2020.

Introduce Entry Financing

High price sensitivity signals an affordability barrier. Partnering with African financial institutions to offer low-deposit financing could unlock a significant untapped buyer segment.

Investigate Post-2019 Drop

The sharpest volume declines occurred from 2020 onward. A deeper analysis factoring in GDP growth, fuel prices, and competitor activity in key African markets is recommended.

08 | Executive Summary

This project demonstrates the ability to translate a real-world business problem into structured SQL queries — progressing from basic filtering and stored procedures to advanced window functions that produce actionable business intelligence.

The two core analyses — Revenue Trend Analysis and Price Elasticity of Demand — answer the two most important questions any automotive regional team needs: Are we pricing right? And which models are truly worth investing in?

Revenue Trend Analysis

Identifies top-performing models and tracks how BMW Africa's revenue has grown or declined across 14 years.

Price Elasticity of Demand

Reveals how sensitive African BMW buyers are to price changes — enabling smarter, data-driven pricing decisions.

09 | Table Created View

```sql
-- Price Elasticity of Demand for BMW X-Class Models (2010-2024)
SELECT
    model,
    year,
    price_usd,
    sales_volume,
    lag(price_usd) over (partition by model order by year) AS prev_price,
    lag(sales_volume) over (partition by model order by year) as prev_sales_volume,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

model	year	price_usd	sales_volume	prev_price	prev_sales_volume	price_change_percentage	volume_change_percentage
3 Series	2010	83300	2746	NULL	NULL	NULL	NULL
3 Series	2010	32861	9975	83300	2746	-60.55	263.26
3 Series	2010	115046	3734	32861	9975	250.10	-62.57
3 Series	2010	105114	8248	115046	3734	-8.63	120.89
3 Series	2012	116827	9454	105114	8248	11.14	14.62
3 Series	2012	81801	3206	116827	9454	-29.98	-66.09
3 Series	2012	42310	2538	81801	3206	-48.28	-20.84
3 Series	2012	101812	8663	42310	2538	140.63	241.33
3 Series	2012	108051	3540	101812	8663	6.13	-59.14
3 Series	2013	80628	9049	108051	3540	-25.38	155.62
3 Series	2013	100262	7895	80628	9049	24.35	-12.75
3 Series	2013	94380	3930	100262	7895	-5.87	-50.22
3 Series	2013	39953	2115	94380	3930	-57.67	-46.18
3 Series	2013	100782	6833	39953	2115	152.25	223.07
3 Series	2014	42600	5783	100782	6833	-57.73	-15.37
3 Series	2014	97715	1482	42600	5783	129.38	-74.37
3 Series	2014	60352	3348	97715	1482	-38.24	125.91

Result 6 | Read Only

```sql
-- Revenue Table on Vehicles with Manual transmitions and Hybrid fuel type in Africa
select *,(price_usd*sales_volume) as total_revenue
from bmw.`bmw sales data (2010-2024)`
where region = 'Africa'
having transmission = 'manual' and fuel_type = 'hybrid'
order by year,total_revenue;
--
-- Helps which Region generates the most revenue which can also highlight business demand
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_KM	Price_USD	Sales_Volume	Sales_Classification	total_revenue
X1	2010	Africa	White	Hybrid	Manual	3.3	194415	62273	344	Low	21421912
X6	2010	Africa	White	Hybrid	Manual	2.3	128175	51235	695	Low	35608325
X5	2010	Africa	Black	Hybrid	Manual	3.3	95833	37994	5110	Low	194149340
3 Series	2010	Africa	Blue	Hybrid	Manual	1.9	43912	83300	2746	Low	228741800
X3	2010	Africa	Grey	Hybrid	Manual	4.2	158043	55462	7748	High	429719576
X1	2010	Africa	White	Hybrid	Manual	3.3	46370	94337	5174	Low	488099638
7 Series	2010	Africa	Grey	Hybrid	Manual	1.7	49503	118028	5144	Low	607136032
M3	2010	Africa	Red	Hybrid	Manual	2	23814	68484	9224	High	631696416
X5	2010	Africa	Blue	Hybrid	Manual	4	67943	119148	6452	Low	768742896
M3	2011	Africa	Blue	Hybrid	Manual	4.3	17779	64280	1002	Low	64408560
M3	2011	Africa	White	Hybrid	Manual	2.9	49603	80312	2792	Low	224231104
X1	2011	Africa	Silver	Hybrid	Manual	4.1	120795	41130	5697	Low	234317610
X3	2011	Africa	Silver	Hybrid	Manual	5	68885	51330	6418	Low	329435940
X6	2012	Africa	Black	Hybrid	Manual	4.6	17763	53320	101	Low	5385320
i8	2012	Africa	Blue	Hybrid	Manual	4	185889	69482	675	Low	46900350
5 Series	2012	Africa	Blue	Hybrid	Manual	3	744	41884	2733	Low	114468972
i8	2012	Africa	Silver	Hybrid	Manual	2.1	81362	38764	5899	Low	228668836

Result 5 | Read Only

Regional revenue table by specification of model i.e transmission and fuel type.

```sql
-- BMW Sales Revenue for each region 2010-2024
select *,(price_usd*sales_volume) as total_revenue
from bmw.`bmw sales data (2010-2024)`
where region = 'Africa'
order by year,total_revenue
;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_KM	Price_USD	Sales_Volume	Sales_Classification	total_revenue
X1	2010	Africa	Grey	Diesel	Manual	2.2	56655	35764	127	Low	4542028
X5	2010	Africa	Red	Diesel	Automatic	4.9	193945	50045	241	Low	12060845
i3	2010	Africa	Red	Electric	Automatic	3.6	195344	111724	147	Low	16423428
5 Series	2010	Africa	Grey	Diesel	Manual	1.7	195818	96809	184	Low	17812856
5 Series	2010	Africa	White	Diesel	Manual	3.6	175955	73328	284	Low	20825152
X1	2010	Africa	White	Hybrid	Manual	3.3	194415	62273	344	Low	21421912
X6	2010	Africa	White	Hybrid	Manual	2.3	128175	51235	695	Low	35608325
7 Series	2010	Africa	Silver	Petrol	Automatic	2.7	63679	67971	615	Low	41802165
5 Series	2010	Africa	Red	Diesel	Manual	3.1	116931	52381	842	Low	44104802
X1	2010	Africa	Grey	Diesel	Automatic	1.6	100516	49598	1252	Low	62096696
X6	2010	Africa	Red	Diesel	Automatic	2.1	94582	37876	1698	Low	64313448
M3	2010	Africa	Red	Electric	Manual	4	15805	45507	1697	Low	77225379
i3	2010	Africa	Blue	Diesel	Manual	1.8	120482	101137	968	Low	97900616
5 Series	2010	Africa	Blue	Hybrid	Automatic	2.4	164245	100115	999	Low	100014885
X5	2010	Africa	Black	Diesel	Automatic	2.8	180321	102317	1021	Low	104465657
i3	2010	Africa	White	Petrol	Manual	3.7	190724	87678	1421	Low	124590438
X1	2010	Africa	Black	Diesel	Automatic	5	37731	33141	5113	Low	169449933

Result 4 | Read Only

Revenue table by region

```sql
call bmw_series_class();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_KM	Price_USD	Sales_Volume	Sales_Classification
3 Series	2012	Middle East	White	Electric	Manual	2.4	101595	117995	500	Low
3 Series	2018	Africa	Black	Electric	Manual	1.8	117061	54762	3293	Low
3 Series	2021	North America	Black	Hybrid	Automatic	3.9	87632	65089	660	Low
3 Series	2023	Europe	Black	Electric	Manual	2.8	128295	86402	4266	Low
3 Series	2019	South America	Silver	Hybrid	Manual	2.8	193021	44950	1703	Low
3 Series	2011	Middle East	Red	Hybrid	Automatic	4	22280	55612	6698	Low
3 Series	2024	South America	Grey	Hybrid	Automatic	3.2	144152	49557	1180	Low
3 Series	2021	Africa	Grey	Diesel	Manual	4.3	170208	40194	4548	Low
3 Series	2010	Asia	Black	Petrol	Manual	2.1	107572	86660	8650	High
3 Series	2020	North America	Silver	Hybrid	Automatic	3	158046	78509	8070	High
3 Series	2018	Asia	White	Hybrid	Automatic	1.7	19899	50651	8157	High
3 Series	2017	Europe	Grey	Petrol	Manual	3.8	89365	86684	9474	High
3 Series	2020	South America	Black	Electric	Automatic	3.8	55451	79774	1142	Low
3 Series	2024	Africa	Grey	Electric	Manual	4.3	69653	113482	4956	Low
3 Series	2019	Africa	Silver	Electric	Manual	4.4	33051	108637	5548	Low
3 Series	2024	South America	Blue	Hybrid	Automatic	2.2	69698	79650	6447	Low
3 Series	2012	South America	Blue	Petrol	Manual	3.7	78678	96429	6974	Low
3 Series	2013	South America	Black	Electric	Automatic	2.3	173100	101189	1476	Low
3 Series	2022	South America	Red	Electric	Automatic	3.6	68978	107968	4129	Low
3 Series	2023	Asia	Grey	Diesel	Manual	2.4	58991	84635	3507	Low
3 Series	2016	Europe	White	Electric	Automatic	4.7	149253	71328	3222	Low
3 Series	2016	Africa	Black	Petrol	Automatic	2.3	189931	71591	6640	Low

Result 1 | Result 2 | Result 3 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_KM	Price_USD	Sales_Volume	Sales_Classification
5 Series	2016	Asia	Red	Petrol	Manual	3.5	151748	98740	8300	High
5 Series	2022	North America	Blue	Petrol	Automatic	4.5	10991	113265	6994	Low
5 Series	2017	Middle East	Silver	Diesel	Manual	1.9	171362	42926	1232	Low
5 Series	2020	Africa	White	Electric	Manual	2.3	163444	119486	4668	Low
5 Series	2017	North America	Silver	Electric	Manual	3.1	57843	111326	4880	Low
5 Series	2024	Asia	Blue	Electric	Manual	3.7	157789	86855	648	Low
5 Series	2011	Middle East	Red	Hybrid	Automatic	3.1	139212	31815	2011	Low
5 Series	2011	Europe	White	Petrol	Automatic	2.1	146963	90224	3683	Low
5 Series	2021	Asia	Blue	Electric	Manual	1.6	140059	82668	271	Low
5 Series	2018	Middle East	Grey	Petrol	Automatic	3.3	45371	87345	1299	Low
5 Series	2019	South America	Red	Hybrid	Manual	4.2	132625	88464	6353	Low
5 Series	2011	Europe	Black	Diesel	Automatic	2.5	116859	103384	9334	High
5 Series	2010	Europe	Red	Petrol	Manual	1.8	194101	34120	1344	Low
5 Series	2016	Europe	Silver	Petrol	Manual	3.2	167770	119412	8291	High
5 Series	2012	Asia	Red	Diesel	Automatic	2.7	124260	39501	6584	Low
5 Series	2024	Middle East	Grey	Petrol	Manual	3.6	120789	49279	9951	High
5 Series	2019	Middle East	Black	Petrol	Automatic	3.8	178723	36571	7403	High
5 Series	2020	South America	Blue	Electric	Manual	1.8	18642	53862	6897	Low
5 Series	2021	Asia	Black	Diesel	Automatic	3.5	133836	38861	9087	High
5 Series	2020	Europe	Blue	Diesel	Manual	3.2	165669	38170	9239	High
5 Series	2017	Europe	Silver	Electric	Manual	3.7	69712	82957	8127	High
5 Series	2016	Middle East	Silver	Petrol	Manual	3.8	135036	38344	8273	High

Result 1 | Result 2 | Result 3 | Read Only

```
14 • call bmw_series_class();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_KM	Price_USD	Sales_Volume	Sales_Classification
7 Series	2020	South America	Black	Diesel	Manual	2.1	122131	49898	3080	Low
7 Series	2020	North America	Silver	Diesel	Automatic	3.8	27403	100015	8111	High
7 Series	2017	North America	Black	Petrol	Automatic	3.2	58288	101791	7521	High
7 Series	2018	Middle East	Blue	Petrol	Manual	4.2	94859	50620	6962	Low
7 Series	2013	Asia	Red	Hybrid	Manual	4.5	144832	118363	1564	Low
7 Series	2012	South America	Silver	Petrol	Manual	3.8	93073	102551	6709	Low
7 Series	2023	North America	Grey	Diesel	Automatic	4.8	64153	117121	4823	Low
7 Series	2024	North America	Grey	Electric	Manual	2.4	80659	92213	5456	Low
7 Series	2024	Africa	Blue	Diesel	Automatic	3.1	132686	42013	8472	High
7 Series	2023	Europe	Red	Electric	Manual	5	50602	83812	2241	Low
7 Series	2016	Africa	Grey	Petrol	Automatic	2.1	49521	93993	8663	High
7 Series	2017	Middle East	Grey	Petrol	Automatic	3	112754	93431	7462	High
7 Series	2022	North America	Silver	Petrol	Automatic	1.6	123623	52099	4212	Low
7 Series	2021	South America	Blue	Petrol	Automatic	1.9	68969	111071	5908	Low
7 Series	2018	Middle East	Black	Electric	Automatic	4.7	38502	85604	5312	Low
7 Series	2017	Europe	White	Petrol	Manual	4.4	185687	106473	8666	High
7 Series	2021	Europe	Black	Diesel	Automatic	4.2	83191	60731	6249	Low
7 Series	2012	North America	Silver	Hybrid	Automatic	4.2	178603	42764	9856	High
7 Series	2022	Middle East	Blue	Diesel	Automatic	2.3	126953	57691	271	Low
7 Series	2018	Asia	Red	Hybrid	Manual	2.5	31798	58752	3606	Low
7 Series	2010	Asia	White	Petrol	Automatic	2.8	28668	65268	2069	Low
7 Series	2018	Africa	Silver	Electric	Manual	4.5	198286	96859	8022	High

Result 1 | Result 2 | Result 3 | Read Only

A set of BMW 3, 5, and 7 Series tables in stored procedure

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